Machine Learning – Professional

Project Proposal

Efficient Image Retrieval Using Hierarchical K-Means Clustering

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The project focuses on enhancing **content-based image retrieval (CBIR)** efficiency through **hierarchical K-means (HKM) clustering**. Traditional CBIR systems rely on exhaustive similarity calculations across massive image databases, which become computationally expensive at scale. By structuring deep visual embeddings into a multi-level tree, HKM enables faster, approximate image retrieval with minimal accuracy loss. Our project plans to replicate the results and analyze the work by *Park and Hwang (2024)*, which demonstrated significant speed improvements while maintaining over 99% retrieval precision across multiple datasets.

The evaluation in the original work used fine-grained visual recognition datasets such as **CARS** and **CUB (Caltech–UCSD Birds)**, which contained diverse object categories and high intra-class variability. These datasets serve as robust benchmarks for assessing both category-level and instance-level retrieval performance. Metrics such as **Recall@1** and **mean average precision (mAP)** will quantify retrieval accuracy, while runtime analysis determine efficiency improvements over baseline models.

Methodologically speaking, the project implements a two-stage pipeline: an **offline clustering phase** that hierarchically organizes image descriptors using K-means, followed by an **online retrieval phase** that traverses the cluster tree using one of three search strategies : Intensive, Auto, or Relaxed, balancing retrieval speed against accuracy. Performance in the original work was compared against state-of-the-art baselines such as CLIP-based UNICOM and CNN-based R-GeM, with the goal of achieving comparable precision at significantly lower computational cost.
The proposed applications expand to real-time visual search tasks such as e-commerce product retrieval, autonomous vehicle perception requiring rapid visual localization, and large-scale surveillance or multimedia indexing systems where efficient and accurate image retrieval is essential.

Use **global descriptors** extracted from pre-trained CNNs (R-GeM) and Vision Transformers (CLIP / ViT-L/14). Each image x_i is represented as a compact feature vector $\mathbf{v}_i \in R^d$.

Then, it deploys Hierarchical K-Means Clustering in two stages.

Offline retrieval stage: recursively cluster database descriptors into a K-ary tree of depth l, minimizing intra-cluster distances.

Online retrieval stage : Given a query descriptor $\mathbf{q}$, traverse the tree using one of three strategies:

- **Intensive Search:** Fastest, single-path traversal.
- **Relaxed Search:** Multiple-node traversal with thresholds s_1, s_2.
- **Auto Search:** Adaptive, threshold-free traversal.

Compute similarity only within candidate leaf nodes to rank retrieved images.

The system's performance will be evaluated using Recall@1 for category-level retrieval and mean average precision (mAP) for object-level retrieval, with additional comparisons focusing on retrieval speed and accuracy retention across different search algorithms.

Reference:

Park, D., & Hwang, Y. (2024). *Efficient Image Retrieval Using Hierarchical K-Means Clustering.* Sensors, 24(8), 2401. https://doi.org/10.3390/s24082401